

Meiru Zhang

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Research areas: Large Language Model, Retrieval-Augmented Generation, Controlled Text Generation, Personalization, Event Extraction and Forecasting

EDUCATION

University of Cambridge, PhD in Language Technology Lab

Oct 2021 – Aug 2026

- Supervisor: Prof. Nigel Collier.
- Scholarship: **Gates Cambridge Scholarship**.
- Research area: Event Extraction and Forecasting, Retrieval-Augmented Generation, LLM Agent.

Imperial College London, MSc in Computing (Machine Learning)

Sept 2018 – Sept 2019

- Honors: **Distinction**
- Master project: **Multimodal Active Learning from Human Data**: applied learning-to-learn technique to tackle data scarcity and predict the engagement level of children with autism during clinical sessions.
- Research area: Personalization, Active Learning, Reinforcement Learning

University College London, BEng in Electronic and Electrical Engineering

Sept 2015 – Jun 2018

- Honors: **Dean's List**, Goldsmith Sessional Prize (2017 and 2016), **GPA: 81 (Top 5%)**.
- Final year project: **Big Data Applied on Registered Patents**: Applied TF-IDF and machine learning to predict the growth rate of fuel cell technologies using published patents.

RESEARCH EXPERIENCE

The Power of Simplicity in LLM-Based Event Forecasting

July 2024 – Mar 2025

- **Outcome**: Investigates how to make LLM-based event forecasting both more accurate and far more economical by replacing an iterative reasoning agent (ReAct) with a single-step retrieval-augmented generation (RAG) pipeline.
- **Findings**: Our experiments reveal that compact, structured statistical distributions and event code lists consistently outperform verbose, unstructured news headlines. Provide insights and guidance toward more reliable, cost-effective forecasting.
- **Skills**: **LLM Agents**, **Agentic RAG**

Can We Instruct LLMs to Compensate for Position Bias?

Dec 2023 – May 2024

- **Outcome**: Proposed **Attention Instruction** to interpret and mitigate the position bias of LLMs through instruction prompting. Visualized the connection between semantic context and attention scores of LLMs.
- **Findings**: We designed instructions to mitigate the position bias problem in LLMs' in-context reasoning. Our experiments show that 1) LLMs do not show to have relative position awareness, 2) LLMs can follow the instruction to attend on a segment, 3) Relative regional attention control can be achieved through repeating position words.
- **Skills**: **Mitigating Position Bias**, Attention Mechanism, **Retrieval-Augmented Generation**

Contrastive Oracle-free Framework for Event Extraction (COFFEE)

May 2022 – Jan 2023

- **Outcome**: Introduced the Oracle-Free Event Extraction (OFEE) task to overcome the limitations of existing event extraction techniques. Developed the novel COFFEE framework.
- **Responsibility**: Directed the research project, analyzed state-of-the-art event extraction approaches, and guided the project towards publication.
- **Skills**: **Event Extraction**, **Controlled Text Generation**, Finetuning.

Dynamic Knowledge Infusion with Reinforcement Learning

Nov 2021 – Mar 2022

- **Objective**: Developed a reinforcement learning-based active learning strategy to dynamically partition the knowledge graph for effective knowledge infusion into pre-trained language models.
- **Responsibility**: Led the project, designed and trained reinforcement learning algorithms capable of adapting to non-stationary environments.

- **Skills:** DQN, Multi-armed Bandit, Active Learning, Continual Learning, Adapter-Tuning

Multi-modal Active Learning from Human Data

Mar 2019 – Aug 2019

- **Outcome:** Developed multimodal models of user states and applied deep reinforcement learning to devise an optimal policy for the active selection of user data.
- **Responsibility:** Created an active learning agent using reinforcement learning (DQN) to select the most informative training samples for a multi-modal multi-class classification task.

PUBLICATIONS

- **DRIFT: Decompose, Retrieve, Illustrate, then Formalize Theorems** Under review. Meiru Zhang, Philipp Borchert, Milan Gritta, and Gerasimos Lampouras.
- **The Power of Simplicity in LLM-Based Event Forecasting** In Proceedings of the First Workshop for Research on Agent Language Models (REALM) at ACL 2025. Meiru Zhang, Auss Abboud, Zaiqiao Meng, and Nigel Collier.
- **Can We Instruct LLMs to Compensate for Position Bias?** In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing. Meiru Zhang, Zaiqiao Meng, and Nigel Collier.
- **COFFEE: A Contrastive Oracle-Free Framework for Event Extraction.** In Proceedings of the First Workshop on Matching From Unstructured and Structured Data (MATCHING 2023, ACL). Meiru Zhang, Yixuan Su, Zaiqiao Meng, Zihao Fu, and Nigel Collier.
- **Multi-modal Active Learning From Human Data: A Deep Reinforcement Learning Approach.** In 2019 International Conference on Multimodal Interaction (ICMI '19). Ognjen Rudovic, Meiru Zhang, Bjorn Schuller, and Rosalind Picard. 2019.

WORK EXPERIENCE

Huawei Noah's Ark Lab, London, UK, Full-time Research Intern in NLP

Feb 2025 – Sept 2025

- Identified the challenges of current retrieval-augmented theorem autoformalization with LLMs as: 1) complex and under-specified query for retrieval and 2) lack of demonstrations for accurate usage of retrieved premises.
- Introduced "Decompose" module to break down the information statements into sub-queries to achieve atomic, precise dependent premise retrieval.
- Integrated "Illustrate" module that selects theorems using the retrieved premises to demonstrate the usage of those premises in a theorem.
- Finetuned retriever model BGE-M3 for dependency retrieval.

Antobot, Computer Vision Engineer

Oct 2018 – Sept 2021

- Constructed instance segmentation dataset: collect videos and pre-process for annotations, construct annotation guidance, review and check annotations.
- Trained and implemented MobileNet and YOLOv5 based models to detect and track fruits end-to-end.
- Built regression model for yield prediction, multi-object tracking and fruit size estimation.
- Integrated the model to edge devices using TensorRT and onnx.

Westlake University, Research Assistant

Sept 2020 – Jan 2021

- Served in the Machine Intelligence Lab (MiLab) organized by Dr. Donglin Wang, conducting in-depth research on machine learning with limited data, researched on combine meta-learning and "learning to active learn" for fast adaptation in non-stationary environments

LANGUAGE & SKILLS

Languages: Chinese (Native), English (Fluent), German (A2)

Programming Languages: Python, Java, Matlab, C, LaTeX

Frameworks/Tools: PyTorch, TensorFlow, Transformers, OpenRL, OpenCV, vLLM, Weights and Biases.